

How far does a dropped sphere land from the point directly beneath its release position?

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Abstract

We analyse the horizontal displacement of a sphere that is released from rest in the rotating-Earth frame. Starting from the full Newton–Coriolis equation, we first derive a compact order-of-magnitude estimate that clarifies which terms are relevant for typical laboratory heights. An approximate analytic solution valid for $h \ll R$ is then compared with an exact Kepler-orbit formulation that treats the drop as a short segment of an elliptic trajectory in an inertial frame. Both approaches agree to within $< 0.5\%$ for $h \leq 100$ km. The eastward and southward deflections cross over at a height $h_{\times}(\phi)$ that increases rapidly with latitude: $h_{\times} \approx 3.0$ km at 20° N, 15.0 km at 50° N, and 24.5 km at 80° N. Above 25 km the motion is always east-dominant. The results provide quantitative benchmarks for high-precision drop experiments and for classroom demonstrations of Coriolis effects.

Keywords: Coriolis deflection, free fall, Earth’s rotation, latitude dependence, analytical mechanics

1 Problem

The Earth rotates with angular velocity ω . A point mass is released from rest at height h above the Earth’s surface at latitude ϕ and allowed to fall freely. Neglect air resistance, but retain the Coriolis and centrifugal inertial forces associated with the rotating reference frame. Determine the horizontal displacement of the impact point from the point vertically beneath the release point.

2 Brief solution

As figure 1, we continue using the diagram and coordinate system from the original book. First, define variables: $\vec{r} = (x, y, z)$, $\vec{g} = (0, 0, -g)$, $\vec{\omega} = (0, \omega \cos(\phi), \omega \sin \phi)$, $\vec{R} = (0, 0, R)$ R is Earth’s radius, m is the particle’s mass. The motion equation can be written:

$$m\ddot{\vec{r}} = m\vec{g} - m\vec{\omega} \times (\vec{\omega} \times (\vec{r} + \vec{R})) - 2m\vec{\omega} \times \vec{v} \quad (1)$$

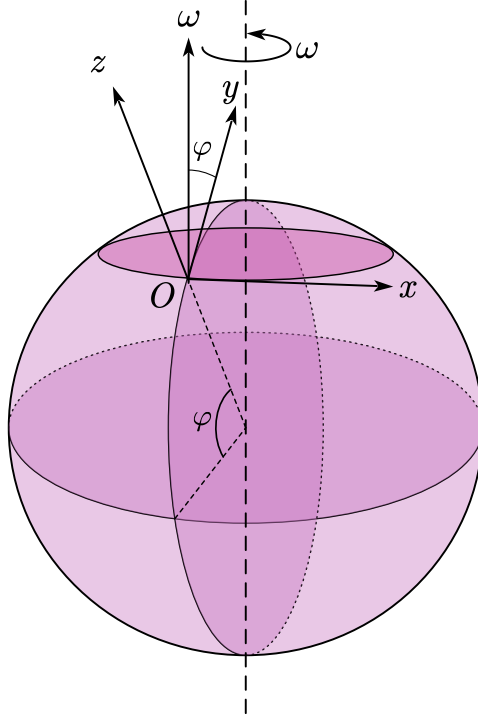


Figure 1: Global Cartesian frame (x, y, z) fixed at the release point: x points east, y north, z vertically upward.

It can be written in component form:

$$\ddot{x} = \omega^2 - 2\omega \cos \phi \dot{z} + 2\omega \sin \phi \ddot{y} \quad (2)$$

$$\ddot{y} = -\omega^2 \cos \phi \sin \phi (z + R) + \omega^2 \sin \phi y - 2\omega \cos \phi \dot{x} \quad (3)$$

$$\ddot{z} = -g + \omega^2 \cos^2 \phi (z + R) - \omega^2 \cos \phi \sin \phi y + 2\omega \cos \phi \dot{x} \quad (4)$$

Performing order-of-magnitude analysis: Since the effect of Earth's rotation is small, we can initially approximate the motion as free fall. Therefore, $x, y, \dot{x}, \dot{y}$ are all small.

The vertical velocity is approximately $\dot{z} \sim \sqrt{2gh} \sim 44$ Use parameters: $R \sim 6 \times 10^6$ $\omega \sim 7 \times 10^{-5}$ $\omega \sim 5 \times 10^{-9}$ However, the term $-\omega^2 \cos \phi \sin \phi R$ cannot be neglected, as its order of magnitude is around 0.03.

The approximate equations:

$$\ddot{x} = 2\omega \cos \phi \dot{z} \quad (5)$$

$$\ddot{y} = -\omega^2 \cos \phi \sin \phi R - 2\omega \cos \phi \dot{x} \quad (6)$$

$$\ddot{z} = -g + \omega^2 \cos^2 \phi R \quad (7)$$

Solving the system of equations yields the result:

$$x = \frac{1}{3}\omega \cos \phi (g - \omega^2 \cos^2 \phi R)t^3 \quad (8)$$

$$y = -\frac{1}{2}\omega \sin \phi (g - \omega^2 \cos^2 \phi R)t^2 \quad (9)$$

$$z = -\frac{1}{2}(g - \omega^2 \cos^2 \phi R)t^2 \quad (10)$$

2.1 Final answer

Substituting the numerical values yields (assuming the particle is released from a height of 100 meters at 40°N latitude):

- Southward deviation: 0.169286 meters
- Eastward deviation: 0.0174786 meters
- Time of flight: 4.517486941 seconds

3 Rigorous solution

As suggested by β , the problem can be solved precisely using celestial mechanics. Specifically, we choose an inertial frame (a cosmic reference frame) in which the Earth rotates with angular velocity ω , and the particle is effectively projected with an initial velocity of $\omega(R + h)$ from a height h above the Earth's surface. At that moment, the point directly below the particle on the Earth's surface is referred to as the ground reference point O . In this reference frame, it is relatively easy to determine the particle's trajectory and the coordinates of the impact point. The complication is that the Earth continues to rotate, so to determine the final position of point O at the time of landing, we must calculate the flight time of the particle. This requires evaluating an integral. While you may opt to numerical solution, we provide a integration method (presented in the appendix at the end).

In this analysis, we consider only the Northern Hemisphere: As shown in the diagram, we first establish three coordinate systems: the Cartesian xyz system, the rotated frame $x^*y^*z^*$, and the spherical coordinate system, where the positive directions are defined as eastward and northward, respectively.

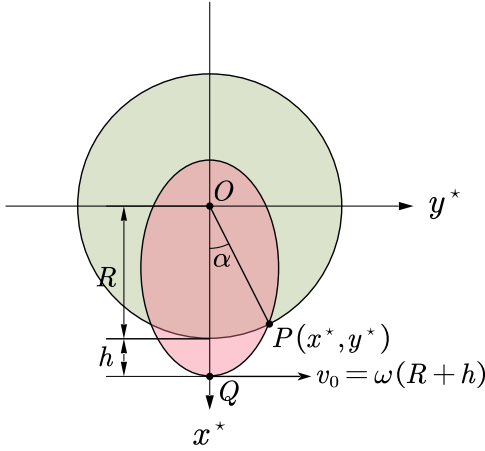


Figure 2: Orbital-plane construction in an inertial (non-rotating) reference frame. The descent is treated as a short arc of an ellipse with perigee distance R and eccentricity e . Point Q denotes the release position, O the origin, and P the impact point.

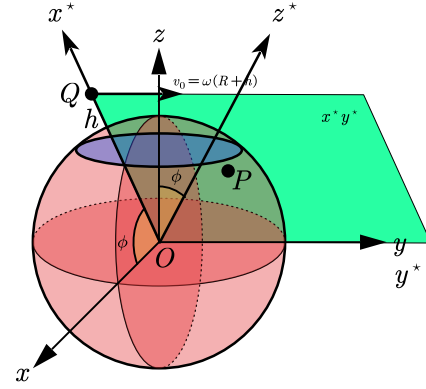


Figure 3: Rotated frame (x^*, y^*, z^*) obtained from the local (x, y, z) frame by a pitch rotation of angle ϕ . Points Q , O , and P mark the release position, the origin, and the impact point, respectively. The particle's subsequent motion is entirely confined to the x^*y^* -plane.

Next, we provide the transformation relations between the rotated coordinate system $x^*y^*z^*$ and the original Cartesian system x, y, z :

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \cos \phi & 0 & -\sin \phi \\ 0 & 1 & 0 \\ \sin \phi & 0 & \cos \phi \end{pmatrix} \begin{pmatrix} x^* \\ y^* \\ z^* \end{pmatrix} = \begin{pmatrix} x^* \cos \phi - z^* \sin \phi \\ y^* \\ x^* \sin \phi + z^* \cos \phi \end{pmatrix} \quad (11)$$

It can be seen that the subsequent trajectory of the particle lies in the plane defined by the x^* and y^* axes.

The energy and angular momentum of the particle are respectively $E = \frac{1}{2}m(R_0^+h)^2\omega^2 \cos^2 \phi$, $L = m(R+h)^2\omega \cos \phi$

Parameters of the orbit:

1. The orbit in system $x^*y^*z^*$: $r = \frac{e}{1+\cos(\pi+\alpha)} = \frac{p}{1-e \cos \alpha}$
2. Orbital semi major axis: $a = \frac{GMm}{-2E} = \frac{GM(R+h)}{2GM-(R+h)^3\omega^2 \cos^2 \phi}$
3. Orbital semi-minor axis: $b = \sqrt{\frac{L^2}{-2mE}} = \sqrt{\frac{(R+h)^5\omega^2 \cos^2 \phi}{2GM-(R+h)^3\omega^2 \cos^2 \phi}}$, $c = \sqrt{a^2 - b^2}$, $e = \frac{c}{a}$, $p = \frac{b^2}{a}$

In Figure 3, the coordinates of the landing point can also be calculated as $P(x^*, y^*, z^*) = (R \cos \alpha_1, R \sin \alpha_1, 0)$.

And the coordinates in the xyz system are $P(x, y, z) = (R \cos \alpha_1 \cos \phi, R \sin \alpha_1, R \cos \alpha_1 \sin \phi)$. The coordinates of point P are converted to spherical coordinates is

$$P_{\text{final}} \left(R, \arctan \left(\frac{\sin \alpha_1}{\cos \alpha_1 \cos \phi} \right), \arctan \left(\frac{\cos \alpha_1 \sin \phi}{\sqrt{\sin^2 \alpha_1 + \cos^2 \alpha_1 \cos^2 \phi}} \right) \right)$$

The definition of α_1 is the angle between the landing point and the initial position (as shown in Figure 3), its value can be obtained by solving the equation $R = \frac{p}{1-e \cos \alpha_1} \Rightarrow \alpha_1 = \arccos(\frac{R-p}{Re})$

Then, determine the time of motion of the particle:

$$\Delta T = \frac{\Delta S(\text{Area})}{\frac{L}{2m}} = \frac{\frac{1}{2} \int_0^{\alpha_1} \left(\frac{p}{1-e \cos \alpha} \right)^2 d\alpha}{\frac{1}{2} (R+h)^2 \omega \cos \phi} \quad (12)$$

$$= \frac{\frac{1}{2} p^2 \cdot \frac{1}{1-e^2} \left(\frac{e \sin \alpha_1}{1-e \cos \alpha_1} + \frac{2}{\sqrt{1-e^2}} \left(\arctan \left(\sqrt{\frac{1+e}{1-e}} \tan \left(\frac{\alpha_1}{2} \right) \right) \right) \right)}{\frac{1}{2} (R+h)^2 \omega \cos \phi} \quad (13)$$

$$= \frac{p^2}{(R+h)^2 \omega \cos \phi} \cdot \frac{1}{1-e^2} \cdot \quad (14)$$

$$\left(\frac{e \sin \alpha_1}{1-e \cos \alpha_1} + \frac{2}{\sqrt{1-e^2}} \left(\arctan \left(\sqrt{\frac{1+e}{1-e}} \tan \left(\frac{\alpha_1}{2} \right) \right) \right) \right) \quad (15)$$

With the time available, the coordinates of the landing reference point at the moment of the particle's landing can be determined. Here, for the convenience of calculation, the spherical coordinates $O_{\text{start}}(R, 0, \phi) \rightarrow O_{\text{final}}(R, \omega \Delta T, \phi)$ are used, and the coordinates of the particle $P_{\text{start}}(R+h, 0, \phi) \rightarrow P_{\text{final}}(R, \theta_{\text{final}}, \phi_{\text{final}}) = \left(R, \arctan \left(\frac{\sin \alpha_1}{\cos \alpha_1 \cos \phi} \right), \arctan \left(\frac{\cos \alpha_1 \sin \phi}{\sqrt{\sin^2 \alpha_1 + \cos^2 \alpha_1 \cos^2 \phi}} \right) \right)$. The difference in latitude and longitude is obtained by subtracting the two coordinate sets; multiplying this angular difference by the appropriate radius gives the linear displacement.

.1 Final answer:

East-west deviation (defined as positive to the east):

$$\begin{aligned} & (\theta_{P_{\text{final}}} - \omega \Delta T) R \cos \phi \\ &= R \cos \phi \left(\arctan \frac{\sin \alpha_1}{\cos \alpha_1 \cos \phi} - \omega \Delta T \right) \\ &= R \cos \phi \left[\arctan \frac{\tan \alpha_1}{\cos \phi} - \frac{p^2}{(R+h)^2 \cos \phi} \frac{1}{1-e^2} \left(\frac{e \sin \alpha_1}{1-e \cos \alpha_1} + \frac{2}{\sqrt{1-e^2}} \arctan \left(\sqrt{\frac{1+e}{1-e}} \tan \left(\frac{\alpha_1}{2} \right) \right) \right) \right] \end{aligned}$$

North-south deviation (defined as positive to the north):

$$(\phi_{P_{\text{final}}} - \phi) R = \left(\arctan \left(\frac{\cos \alpha_1 \sin \phi}{\sqrt{\sin^2 \alpha_1 + \cos^2 \alpha_1 \cos^2 \phi}} \right) - \phi \right) R \quad (16)$$

$$= \left(\arctan \left(\frac{\sin \phi}{\sqrt{\tan^2 \alpha_1 + \cos^2 \phi}} \right) - \phi \right) R \quad (17)$$

with

$$\alpha_1 = \arccos\left(\frac{R-p}{eR}\right) = \arccos\left(\frac{R - \frac{b^2}{a}}{R\sqrt{1 - \frac{b^2}{a^2}}}\right) \quad (18)$$

$$= \arccos\left(\frac{R - \frac{b^2}{\frac{GM(h+R)}{2GM - \omega^2(h+R)^3 \cos^2(\phi)}}}{R\sqrt{1 - \frac{\omega^2(h+R)^5 \cos^2(\phi)}{\frac{(2(GM(h+R))) (2GM - \omega^2(h+R)^3 \cos^2(\phi))}{2GM - \omega^2(h+R)^3 \cos^2(\phi)}}}}}\right) \quad (19)$$

$$a = -\frac{GMm}{2e} = \frac{GM(h+R)}{2GM - \omega^2(h+R)^3 \cos^2(\phi)} \quad (20)$$

$$b = \sqrt{-\frac{L^2}{2mE}} = \sqrt{\frac{\omega^2(h+R)^5 \cos^2(\phi)}{2GM - \omega^2(h+R)^3 \cos^2(\phi)}} \quad (21)$$

Using the values $R = 6.3710110^6 \text{m}$, $\omega = \frac{2\pi}{24 \cdot 3600} / \text{rad}$, $M = 5.972210^{24} \text{kg}$ we get (a height of 100 meters at 40°N latitude):

- Southward deviation: 0.1693027346 meters
- Eastward deviation: 0.01676045865 meters
- Time of flight: 4.5176386115 seconds

.2 What happens if the values of h and ϕ are different?

The results for additional parameter sets are summarised below. As figure 4,5,6,7,8

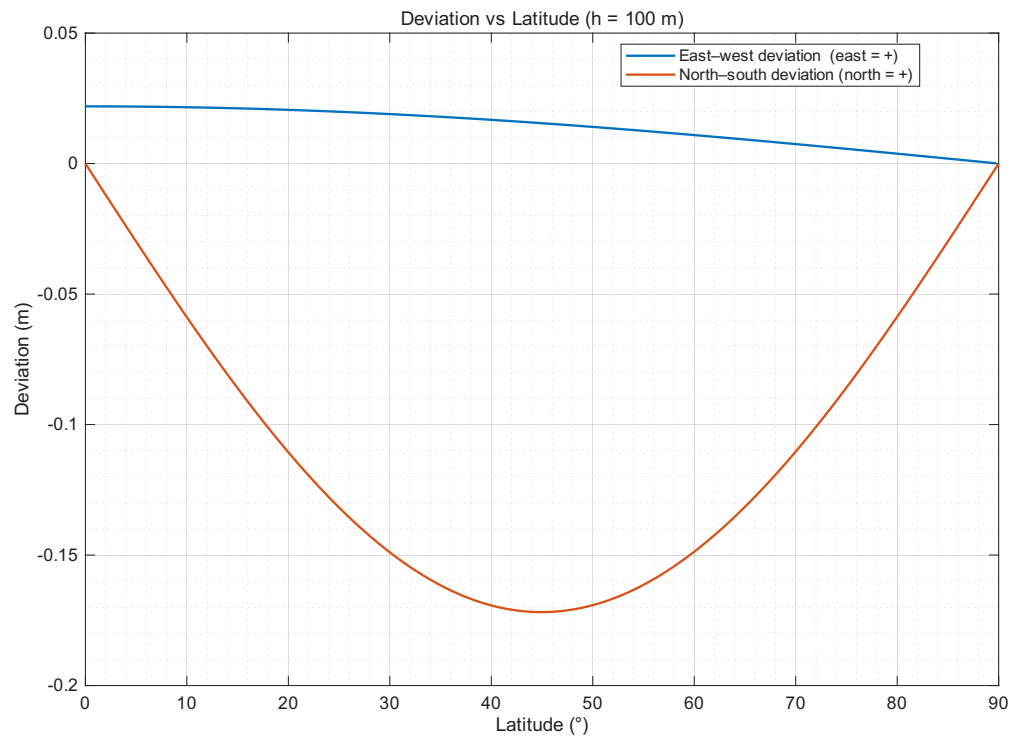


Figure 4: Latitude dependence of the deviation at a release height of 100 m

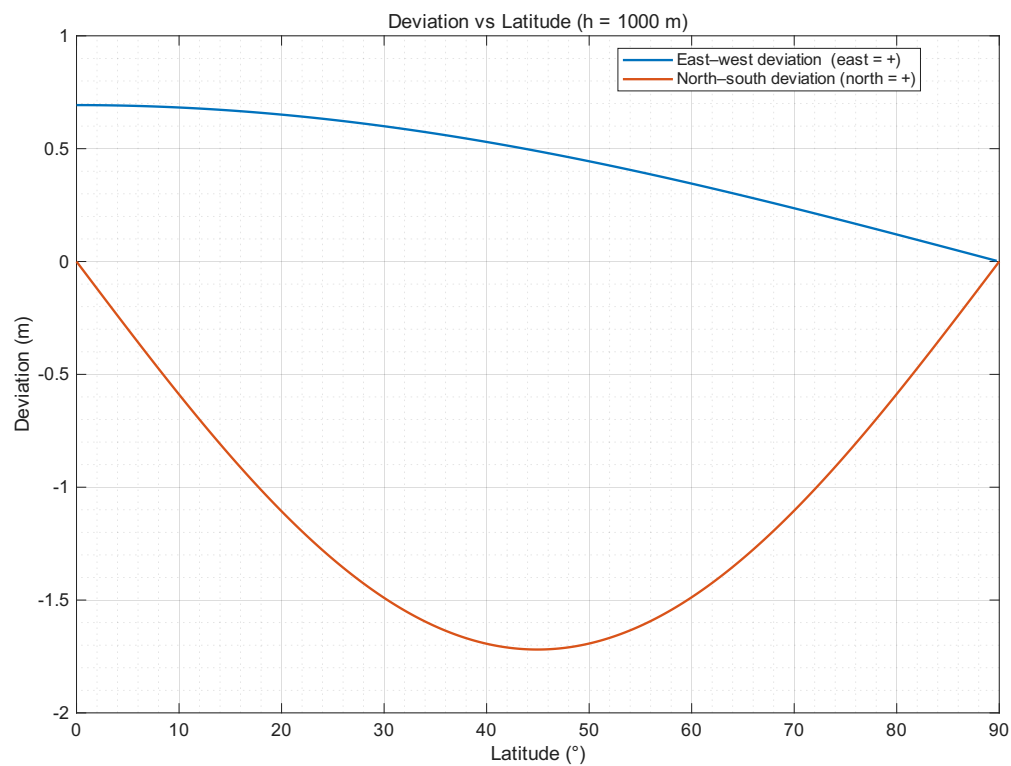


Figure 5: Latitude dependence of the deviation at a release height of 1000 m

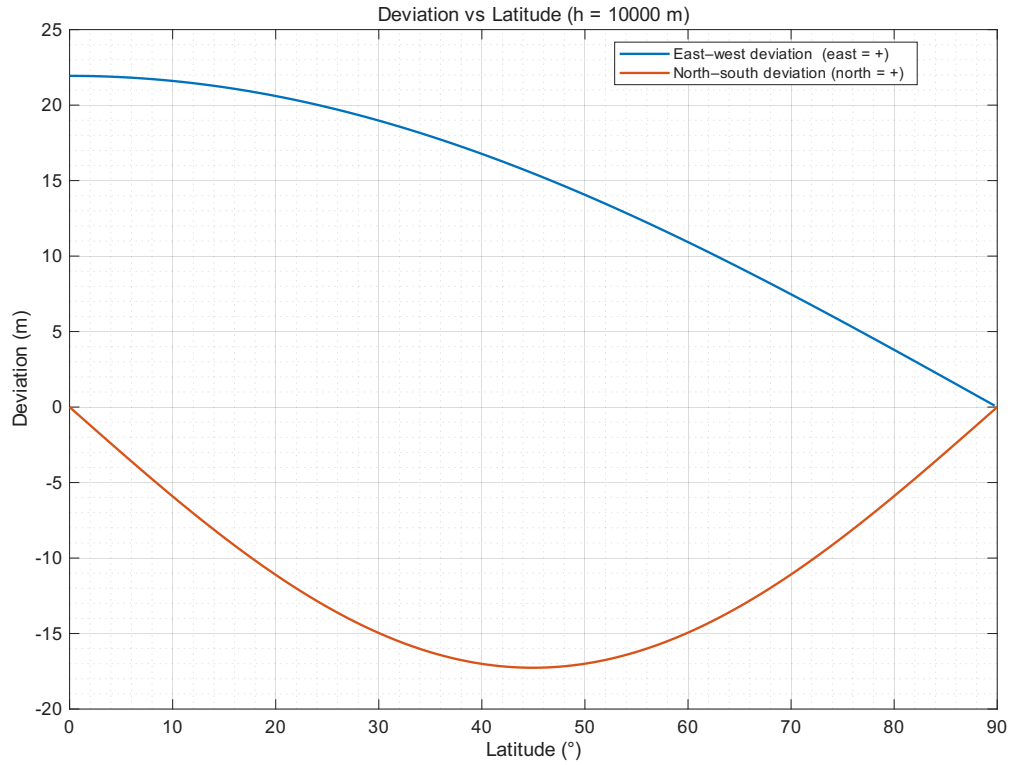


Figure 6: Latitude dependence of the deviation at a release height of 10000 m

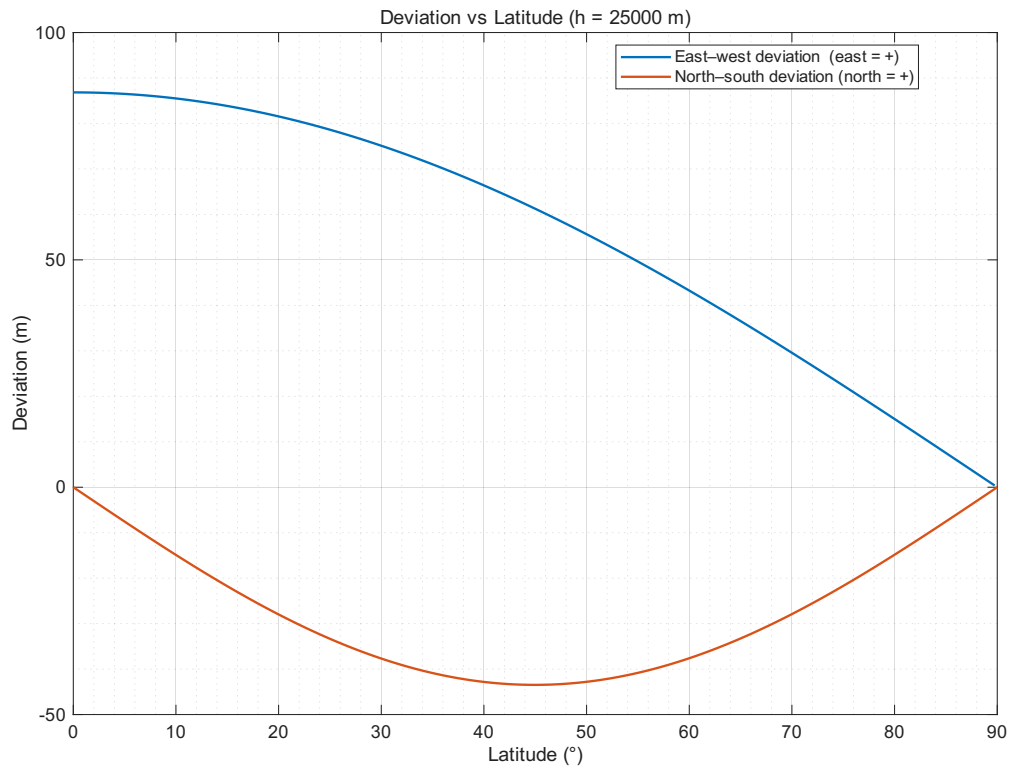


Figure 7: Latitude dependence of the deviation at a release height of 25000 m

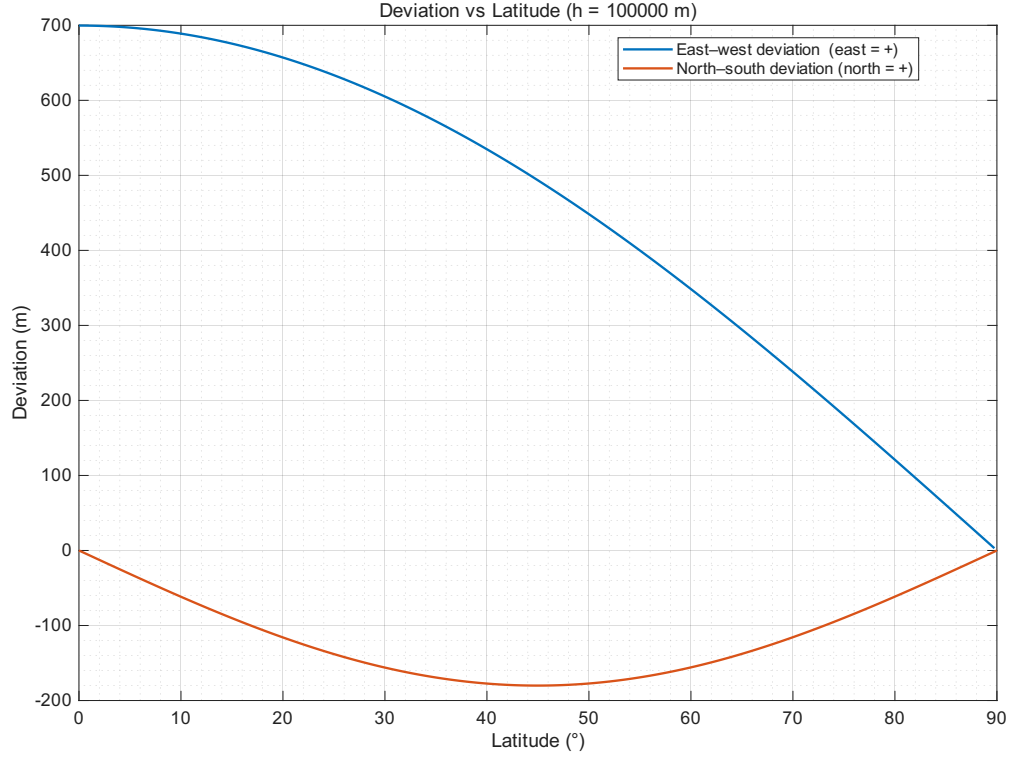


Figure 8: Latitude dependence of the deviation at a release height of 100000 m

.3 Eastward vs. Southward Deviation: Key Drivers

We define $f(h, \phi)$ as the east–west deviation (positive eastward) and $g(h, \phi)$ as the north–south deviation (positive northward). To obtain an initial overview of their behaviour, we tabulate numerical values of f and g for several latitudes ϕ . Each row of the table lists, in sequence, the release height, $f(h, \phi)$, $g(h, \phi)$, and the sum $f(h, \phi) + g(h, \phi)$.

Table 1: Sample heights around the root of $f(h, \phi) + g(h, \phi) = 0$ (values in metres)

(a) $\phi = 20^\circ\text{N}$				(b) $\phi = 50^\circ\text{N}$				(c) $\phi = 80^\circ\text{N}$			
Height	$f(h, \phi)$	$g(h, \phi)$	$f + g$	Height	$f(h, \phi)$	$g(h, \phi)$	$f + g$	Height	$f(h, \phi)$	$g(h, \phi)$	$f + g$
1	0.00	-0.00	-0.00	1	0.00	-0.00	-0.00	1	0.00	-0.00	-0.00
1 000	0.65	-1.11	-0.46	1 000	0.44	-1.69	-1.25	1 000	0.12	-0.59	-0.47
1 894	1.70	-2.10	-0.40	13 671.5	22.48	-23.28	-0.80	23 459.5	13.64	-13.92	-0.28
2 894	3.21	-3.21	0.00	14 671.5	24.99	-24.99	0.00	24 459.5	14.52	-14.52	0.00
3 894	5.00	-4.31	0.69	15 671.5	27.59	-26.71	0.89	25 459.5	15.42	-15.12	0.30
25 500	84.01	-28.55	55.47	25 500	57.33	-43.66	13.67	25 500	15.46	-15.14	0.31

The sign of $f(h, \phi) + g(h, \phi)$ reveals which contribution—eastward or southward—dominates the overall

deflection.¹ Accordingly, we solve the balance equation

$$f(h, \phi) + g(h, \phi) = 0$$

to obtain an implicit function $h = h(\phi)$. This curve (illustrated below) marks the locus where the eastward and southward deviations are equal in magnitude.

- **Below** the curve (shaded area) the southward deflection is larger, consistent with the numerical tables given earlier.
- **Above** the curve the eastward deflection predominates.
- In particular, for heights exceeding 25 000 m the eastward component is dominant at all latitudes considered.

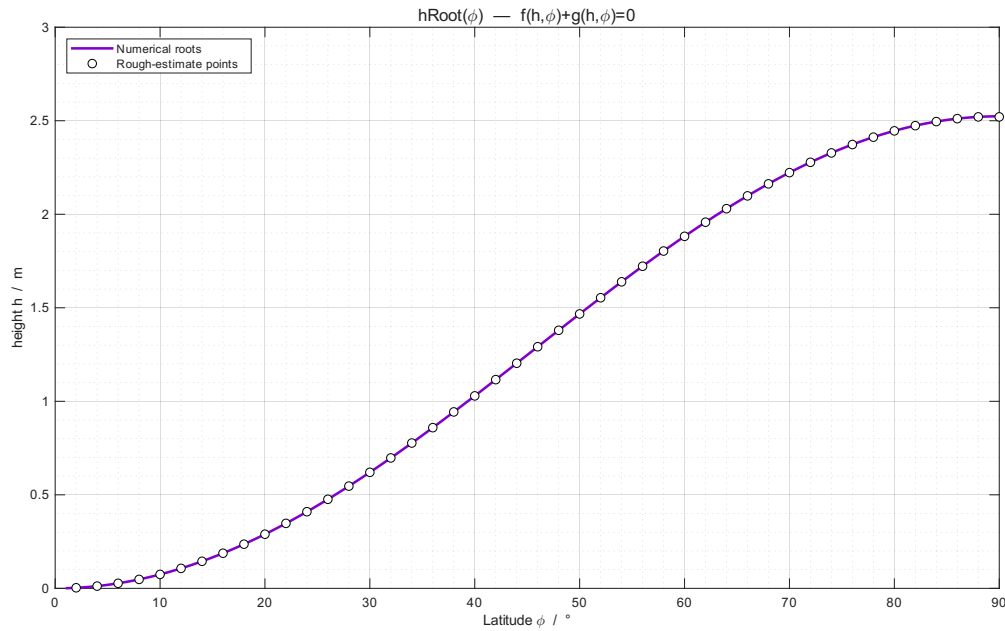


Figure 9: Phase diagram indicating whether the eastward or the southward deflection is dominant

.4 Ground-level footprints

Finally, we examine the trajectories recorded on the ground when a series of spheres is released from different heights at various latitudes. The resulting ground-track patterns are presented in the following figure.

¹Positive values indicate a larger eastward component, whereas negative values imply a larger southward component.

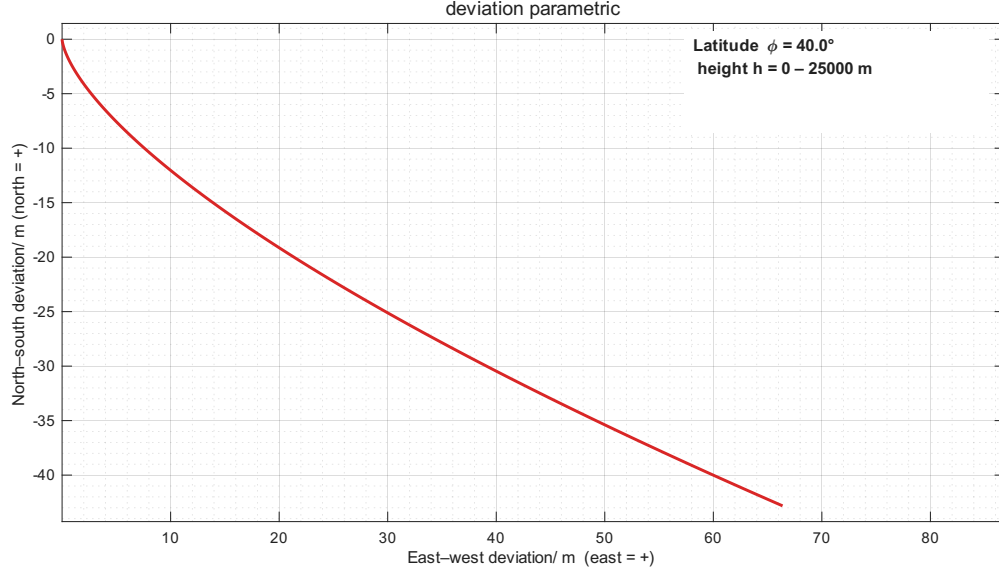


Figure 10: Ground-level tracks of spheres released from rest (Earth-fixed frame) at successive altitudes

A Integrator process

(Feynman's technique of integration - differentiating under the integral sign) Let $I_K = \int \frac{1}{(e \cos(\alpha) + 1)^K} d\alpha$
We can get

$$\begin{aligned}
 I_1 &= \int \frac{1}{e \cos(\alpha) + 1} d\alpha \\
 &= \int \frac{2 \left(\sin^2 \left(\frac{\alpha}{2} \right) + \cos^2 \left(\frac{\alpha}{2} \right) \right)}{\sin^2 \left(\frac{\alpha}{2} \right) + \cos^2 \left(\frac{\alpha}{2} \right) + 2e \left(\cos^2 \left(\frac{\alpha}{2} \right) - \sin^2 \left(\frac{\alpha}{2} \right) \right)} d\frac{\alpha}{2} \\
 &= \int \frac{2 \left(\tan^2 \left(\frac{\alpha}{2} \right) + 1 \right)}{\tan^2 \left(\frac{\alpha}{2} \right) + 2e \left(1 - \tan^2 \left(\frac{\alpha}{2} \right) \right) + 1} d\frac{\alpha}{2} \\
 &= 2 \int \frac{1}{(1 - e) \tan^2 \left(\frac{\alpha}{2} \right) + e + 1} d \tan \left(\frac{\alpha}{2} \right) \\
 &= \frac{2 \arctan \left(\sqrt{\frac{1-e}{e+1}} \tan \left(\frac{\alpha}{2} \right) \right)}{\sqrt{1 - e^2}} \\
 \frac{dI_K}{de} &= \int \frac{\partial}{\partial e} \frac{1}{(e \cos(\alpha) + 1)^K} d\alpha \\
 &= \text{(When the limits of integration are independent of } e \text{)} \\
 &= - \frac{k \int \frac{e \cos(\alpha) + 1}{(e \cos(\alpha) + 1)^{K+1}} d\alpha}{e} \\
 &= - \frac{k \left(\int \frac{1}{(e \cos(\alpha) + 1)^K} d\alpha - \int \frac{1}{(e \cos(\alpha) + 1)^{K+1}} d\alpha \right)}{e} \\
 &= - \frac{k (I_K - I_{K+1})}{e}
 \end{aligned}$$

So $I_{K+1} = \frac{e}{k} \frac{dI_K}{de} + I_K$

Once the recurrence relation is given, each subsequent term can be calculated.

Here we present the I_2 :

$$\begin{aligned}
I_2 &= e \frac{dI_1}{de} + I_1 \\
&= \frac{2 \arctan \left(\sqrt{\frac{1-e}{e+1}} \tan \left(\frac{\alpha}{2} \right) \right)}{\sqrt{1-e^2}} \\
&\quad + e \left(\frac{2e \arctan \left(\sqrt{\frac{1-e}{e+1}} \tan \left(\frac{\alpha}{2} \right) \right)}{(1-e^2)^{3/2}} + \frac{\left(-\frac{1-e}{(e+1)^2} - \frac{1}{e+1} \right) \tan \left(\frac{\alpha}{2} \right)}{\sqrt{\frac{1-e}{e+1}} \sqrt{1-e^2} \left(\frac{(1-e) \tan^2 \left(\frac{\alpha}{2} \right)}{e+1} + 1 \right)} \right) \\
&= \frac{2 \arctan \left(\sqrt{\frac{1-e}{e+1}} \tan \left(\frac{\alpha}{2} \right) \right)}{\sqrt{1-e^2}} \\
&\quad + e \left(\frac{2e \arctan \left(\frac{(1-e) \tan \left(\frac{\alpha}{2} \right)}{\sqrt{1-e^2}} \right)}{(1-e^2)^{3/2}} + \frac{2 \left(\frac{(1-e)e \tan \left(\frac{\alpha}{2} \right)}{(1-e^2)^{3/2}} - \frac{\tan \left(\frac{\alpha}{2} \right)}{\sqrt{1-e^2}} \right)}{\sqrt{1-e^2} \left(\frac{(1-e)^2 \tan^2 \left(\frac{\alpha}{2} \right)}{1-e^2} + 1 \right)} \right) \\
&= \frac{2\sqrt{1-e^2} \arctan \left(\frac{\sqrt{1-e} \tan \left(\frac{\alpha}{2} \right)}{\sqrt{e+1}} \right) + \frac{e(e^2-1) \sin(\alpha)}{e \cos(\alpha)+1}}{(1-e^2)^2} \\
&= \frac{\frac{2 \arctan \left(\sqrt{\frac{1-e}{e+1}} \tan \left(\frac{\alpha_1}{2} \right) \right)}{\sqrt{1-e^2}} - \frac{e \sin(\alpha_1)}{e \cos(\alpha_1)+1}}{1-e^2}
\end{aligned}$$

Similarly, the following items can be obtained.